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B.Tech. Degree V Semester Examination November 2018**ME 15-1502 MECHANICS OF MACHINERY**
(2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Explain steering gear mechanism with a neat figure.
(b) Derive an expression for the magnitude of Coriolis component of acceleration.
(c) Explain the working of a pantograph with a neat sketch.
(d) Explain the different types of gear trains.
(e) Compare cycloidal and involute gear tooth profiles.
(f) Derive an expression for displacement of a tangent cam operating on a radial-translating roller follower, when the contact is on the nose.
(g) Derive an expression for the efficiency of a screw jack.
(h) Obtain the Freudenstein's equation for a four bar mechanism.
(i) Explain the working of a centrifugal clutch with a neat figure.
(j) Explain the working of a rope brake dynamometer with a neat figure.

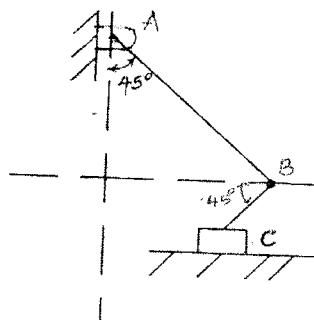
PART B

(4 × 10 = 40)

- II. (a) Derive an expression for the ratio of shaft velocities for a Hooke's joint. (4)
(b) In a slider crank mechanism, the length of crank OB and connecting rod AB are 125 mm and 500 mm respectively. The centre of gravity G of the connecting rod is 275 mm from the slider A. The crank speed is 600 rpm clockwise. When the crank has turned 45° from the inner dead centre, determine (i) velocity of the slider A (ii) velocity of the point G (iii) angular velocity of the connecting rod AB. (6)

OR

- III. In the mechanism shown in figure, the slider C is moving to the left with a velocity of 1.0 m/s and an acceleration of 2.5 m/s². The dimensions of various links are AB = 3 m inclined at 45° with the vertical and BC = 1.5 m inclined at 45° with the horizontal. Determine the angular acceleration of the links AB and BC. (10)



(P.T.O.)

- IV. A 20° involute pinion with 20 teeth drives a gear having 60 teeth. Module is 8 mm and addendum of each gear is 10 mm. (i) State whether interference occurs or not. Give reasons. (ii) Find the length of path of approach and arc of approach, if pinion is the driver.

OR

- V. In a reverted epicyclic gear train, the arm A carries two wheels B and C and a compound wheel D-E. The wheel B gears with E and the wheel C gears with wheel D. The number of teeth on wheels B, C and D are 150, 60 and 180 respectively. Determine the speed and direction of wheel C, when B is fixed and the arm A makes 200 rpm counter clockwise. (10)

- VI. In a symmetrical tangent cam operating a roller follower, the least radius of cam is 30 mm and roller radius is 17.5 mm. The angle of ascent is 75° and the total lift is 17.5 mm. The speed of the cam shaft is 600 rpm. Calculate: (i) The principal dimensions of the cam (ii) The acceleration of the follower at the beginning of lift, where straight flank merges into the circular nose and at the apex of the circular nose. Assume that there is no dwell between ascent and descent. (10)

OR

- VII. A conical pivot bearing supports a vertical shaft of 200 mm diameter. It is subjected to a load of 30 kN. The angle of the cone is 120° and the coefficient of friction is 0.025. Find the power lost in friction when the speed is 140 rpm, assuming (i) uniform pressure condition (ii) uniform wear condition. (10)

- VIII. For the four bar linkage, the following data are given: $\theta = 60^\circ$, $\phi = 90^\circ$, $\omega_{IP} = 3 \text{ rad/s}$, $\omega_{OP} = 2 \text{ rad/s}$, $\alpha_{IP} = -1 \text{ rad/s}^2$, $\alpha_{OP} = 0 \text{ rad/s}^2$. Determine the link-length ratios. Sketch the mechanism taking the length of the shortest link as 30 mm. θ and ϕ are the input crank angle and output follower angle respectively. (10)

OR

- IX. In a belt transmission dynamometer, the driving pulley rotates at 300 rpm. The distance between the centre of the driving pulley and the dead mass is 800 mm. The diameter of each of the driving as well as the intermediate pulleys is equal to 360 mm. Find the value of dead mass required to maintain the lever in a horizontal position when the power transmitted is 3 kW. Also find its value when the belt just begins to slip on the driving pulley, μ being 0.25 and the maximum tension in the belt 1200 N. (10)

